Disability and the Built Environment

Anwar Baroudi

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- Undergrad at UC Berkeley, CS
- Current MCP (possibly + MTE)
- Legally Blind (achromatopsia)
- Grew up in Mississippi (sad)
- Pedestrian safety and disability advocate

Overview

Topics

- What is Disability
- What is Accessibility
- Learning through the Built Environment
- Data, and it’s challenges
- Applications

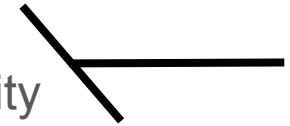
Key Theme

- Importance of Qualitative methods and understanding
  - That is, talk to people with disabilities!
- Power of Universal Design

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Definitions are essential!!

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What is Disability,
Who has them, and
What are the Stakes?

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Who has them,
What are the Stakes?

This question does not have an easy answer. Stigma and culture make answering it even harder.

Some Government Definitions (because there are multiple)

ADA: legal definition that includes people who have a record or regard of a disability

CDC: medical definition focused more on individual impairment and less on societal factors

Generally, refers to folks that have some impairment that limits important activities

What is Disability,
Who has them,
What are the Stakes?

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Government estimates differ depending on who you ask,

- ACS (census) says 13%
- CDC says 28%
- Social Security has a less reliable snapshot, claims 1 in 3 young *workers* will die or qualify for SSDI before reaching retirement age

These don’t generally count children, or elders who don’t claim to have a disability

What is Disability,
Who has them,
What are the Stakes?

This question has an easy answer.

Not accommodating people with disabilities leads to our disenfranchisement, to a total loss in autonomy, an inability to participate in society, and for many, to death

What is Disability,
Who has them,
What are the Stakes?



Do these actually matter?

What is Disability,
Who has them,
What are the Stakes?



Do these actually matter?

Yes.

If you are making policy, or doing research, who that effort is for is extremely important.

Walk scores…for whom?

Who is this wayfinding for?

Where should we prioritize making improvements?

What is Disability,
Who has them,
What are the Stakes?



Do these actually matter?

But Also No.

The stakes are plenty high enough that no matter how small the population, we should be considered.

Principle of Universal Design suggests that we are all better off for it too.

About Universal Design

Universal Design (UD) is the design and composition of an environment so that it can be accessed, understood and used to the greatest extent possible by all people regardless of their age, size, ability or disability. An environment (or any building, product, or service in that environment) should be designed to meet the needs of all people who wish to use it. This is not a special requirement, for the benefit of only a minority of the population. It is a fundamental condition of good design.

Universal Design

Design for the most vulnerable and all of us see benefits

Classic example: ramps

> Mandated all over for wheelchair users, end up being extremely useful for strollers, older folks, anyone with temporary injuries, people moving heavy things, etc.

So… how do we measure it?

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What is Accessibility

This question does not have an easy answer. Stigma and culture make answering it even harder.

What is Accessibility

Depends on the subject of description

Generally, when describing a place or service, it’s the quality of being able to be reached, entered, or used regardless of ability

When describing a system or region e.g. an accessible city, it’s that its designed so that people of all abilities can equally engage and participate

When it is the subject; when it’s used as a noun, it’s the practice of successfully designing with people of all abilities in mind.

What is**n’t** Accessibility

Often confused with **mobility**

Mobility is the ability, and magnitude, to move freely

High mobility means you can travel very far easily, does not mean everyone benefits from it equally.

What is**n’t** Accessibility

Often confused with **mobility**

Example of high mobility:

Most of the USA; cars give us incredible distance and flexibility quite easily. If you can drive them

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Example of low mobility:

UC berkeley campus; its pretty small and yet it takes time to cross, you can’t just drive or fly through it

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UC berkeley campus; its pretty small and yet it takes time to cross, you can’t just drive or fly through it

Mobility and Accessibility don’t have to oppose…but in practice they often do. Given the choice, high accessibility + low mobility for dense urban spaces is preferable to high mobility + low accessibility

Ok, so how do we **measure** accessibility

This question does not have an easy answer—

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Specifically lets measure systems/regions

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This question is the topic of decades (centuries?) of academic debate. Some attempts:

Walk Scores

Gravity Measures

Utility-based Measures

Access Ratio

Ok, so how do we **measure** accessibility

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Walk Scores proximity of amenities

Gravity Measures

Utility-based Measures

Access Ratio

Ok, so how do we **measure** accessibility

This question is the topic of decades (centuries?) of academic debate. Some attempts:

Walk Scores

Gravity Measures attractiveness of destinations (that is how they attract each other and thus clump useful things together)

Utility-based Measures

Access Ratio

Ok, so how do we **measure** accessibility

This question is the topic of decades (centuries?) of academic debate. Some attempts:

Walk Scores

Gravity Measures

Utility-based Measures minimizing trip or tour lengths to the most useful places

Access Ratio

Ok, so how do we **measure** accessibility

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Walk Scores

Gravity Measures

Utility-based Measures

Access Ratio $\dfrac{\text{\#of important places within 45 mins reachable without a car}}{\text{\#of important places within 45 mins reachable with a car}}$

Ok, so how do we measure accessibility…**idk**

There is no standard,

There couldn’t be a standard,

And they all kinda suck…

Lesson: Quantitative Methods don’t solve everything

If we want to make progress we need qualitative methods and context.

So let’s get Qualitative!

Lots of ways to accommodate, or fail to accommodate, people with disabilities, let’s focus on the built environment, in particular the public right of way

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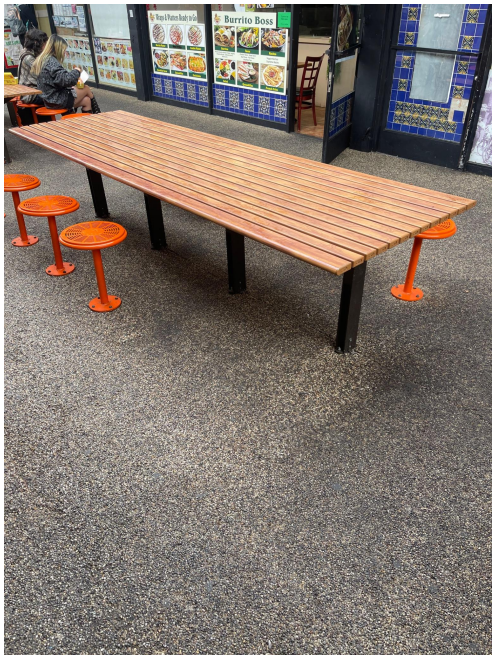
Disclaimer

the way to learn about and accommodate people with disabilities is to talk to and work with people with disabilities, these are complex issues and no one person has all the answers, especially for experiences they haven’t lived.

That includes me.

I have personal experience with low vision, and alot of exposure and experience working with folks with vision, mobility and to some extent hearing. I’ll be talking about those but it will not be exhaustive.

We’ll come back to this…



Ever noticed this empty space at the corner of this table?

Mobility challenges

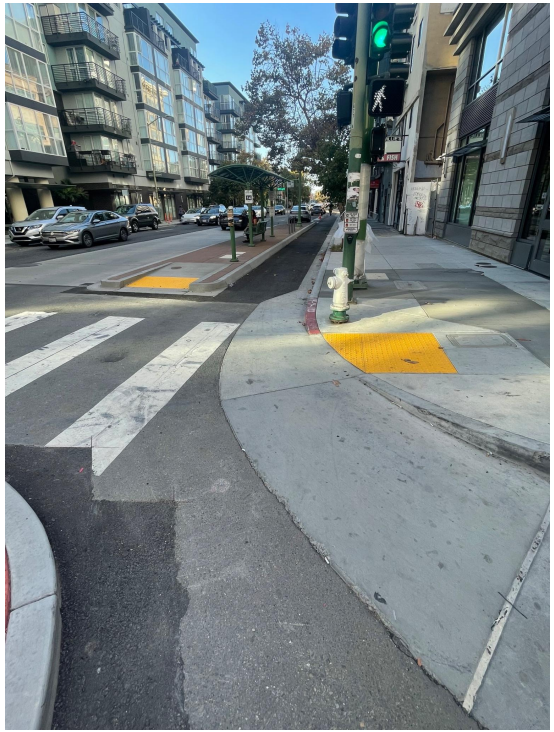
Talking primarily about people who use mobility assistive technology, wheelchairs/powerchairs, walkers, etc.

Important considerations

People with mobility impairments usually need **level ground** or as close to it as can get.

Hills are hard, cracks/gaps are dangerous, curbs and similar bumps are impassible.

Mobility challenges: Ramps



Fundamental piece of infrastructure that connects the curb to the street, absolutely necessary at every crossing.

Come in two main flavors, bi-directional (pictured, good) and uni-directional (bad)

Other niche flavors here and there, usually not great for some reason or other

Mobility challenges: Ramps



Example of a Uni-directional ramp

They were here purely to save money or because there was some storm drain or similar obstacle to using bi-directional ramps. (“unfeasible”)

Mobility challenges: Ramps



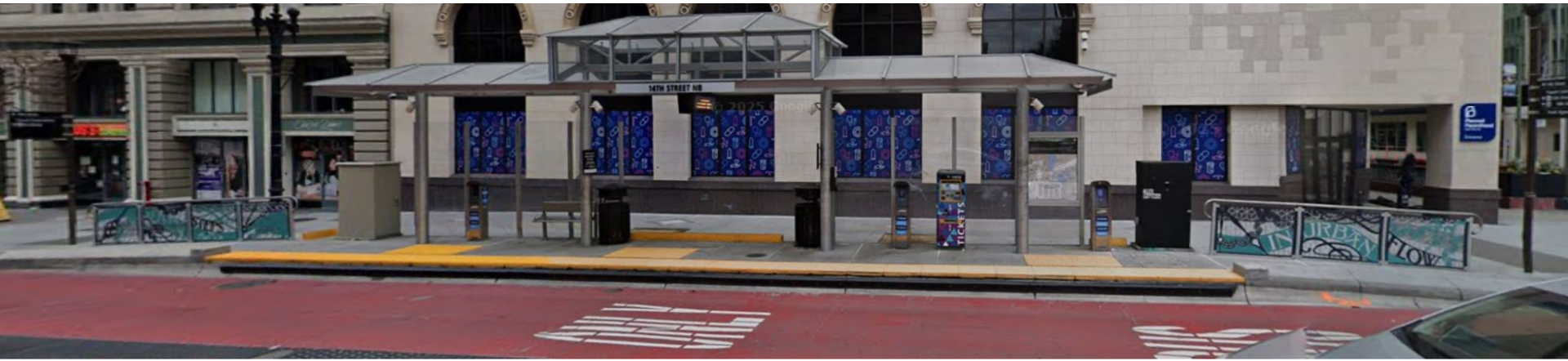
Example of a Uni-directional ramp

They were here purely to save money or because there was some storm drain or similar obstacle to using bi-directional ramps. (“unfeasible”)

Ramps don’t just provide connectivity, they provide direction, and this type of ramp throws you out right into the middle of the crosswalk!

Bad for users of the ramp, and also for drivers who need to understand what peds are doing

Mobility challenges: Level Boarding



Bus platform is extra high, so that wheelchair users can wheel straight onto the bus, no additional ramps needed!

Mobility challenges: Seating



What’s wrong with this? Look, two benches!

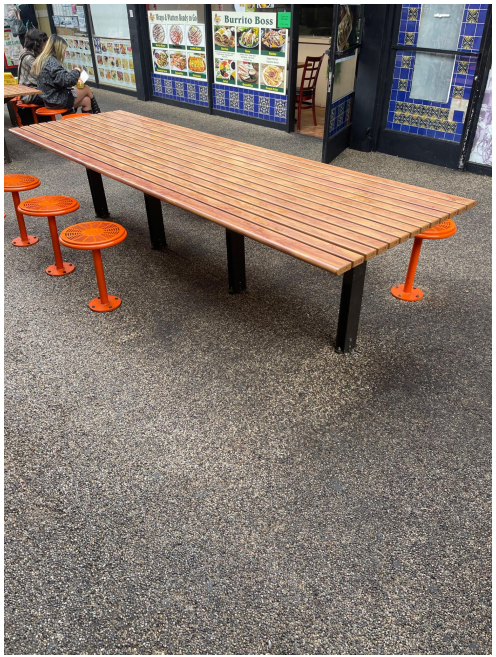
Mobility challenges: Seating



What's wrong with this? Look two benches!

…unless you can't use benches. Hmm if it's raining or too sunny, where do you go to get shade if you are in a wheelchair?

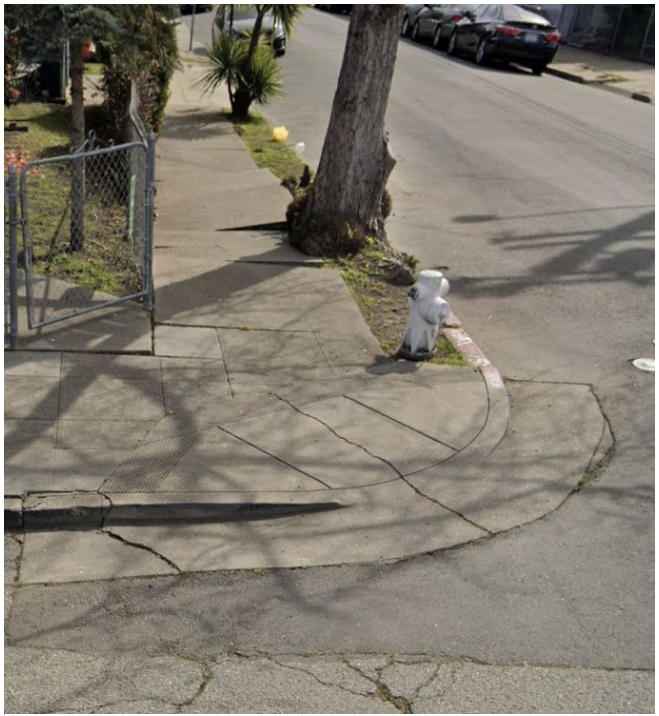
Mobility challenges: Seating



Ever wondered what this empty space at the corner of this table is for?

Do you wonder what it’s for now?

Mobility challenges: Sidewalks



Uprooted, cracked, and otherwise damaged sidewalks are often a dead end for wheelchair users, not to mention major trip hazards for everyone else

Obstructions to the sidewalk can be structural damage, but often they are also trash or **cars parked on the sidewalk (!!)**

Mobility challenges: Construction



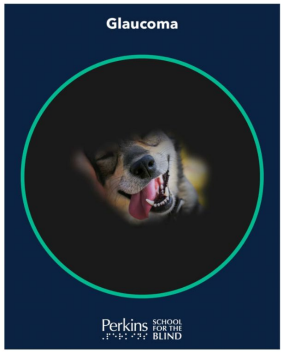
What do you do in a wheelchair here?

(normally there is some tiny awful asphalt heap or plywood plank fashioned as a ramp of to the side…this one actually doesn't even have one of those!)

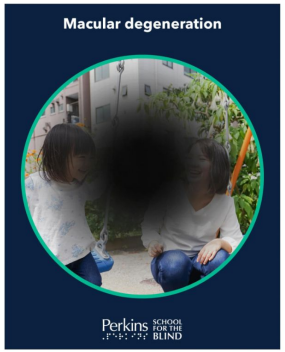
Vision challenges:

Refers to a whole spectrum of low-vision to blind. This also includes deficiencies in color-vision

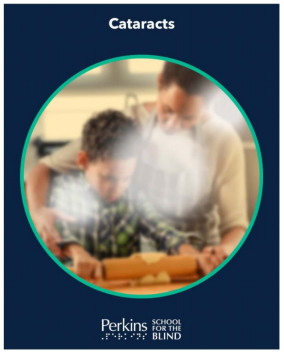
Blindness itself takes many forms, it's not just pitch black for everyone.



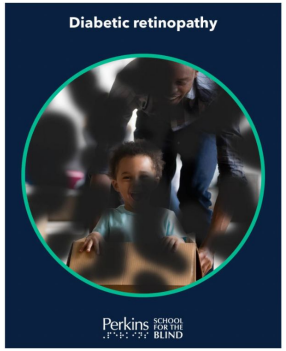
An image of a dog, with much of the surrounding area darkened and blurred, creating a pinhole view, representing the effects of glaucoma.



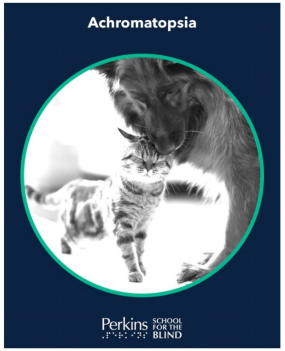
A little girl on a swing next to her mother, with a dark cloud obscuring the middle of the scene, representing the effects of macular degeneration



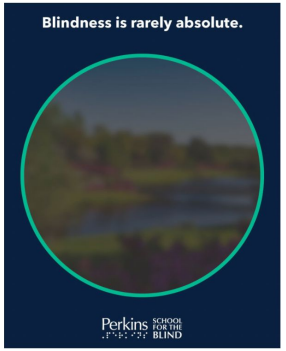
A yellowed scene of a mother and son baking a pie in a kitchen, partially obscured by blurry white spots to represent cataracts.



A scene of a father pushing his laughing young son around in a cardboard box, partially obscured by a series of dark, blurry spots, to represent diabetic retinopathy.



A black and white scene of a dog and a cat snuggling, to represent the complete or partial absence of color vision, also known as achromatopsia.



A dim and blurry scene of a green field, a pond, and trees represents another way that blindness is rarely absolute.

Vision challenges

Refers to a whole spectrum of low-vision to blind.

Blindness itself takes many forms, it’s not just pitch black for everyone.

Important considerations

Visual communication is obviously insufficient, tactile and audio alternatives are needed

Just going in a straight line is very hard, let alone having wider spatial awareness

Vision Challenges: Audio Wayfinding



Pedestrian Push Buttons (often referred to derogatively as Beg Buttons)

- Give audio (and tactile) feedback for when it is time to cross
- Names the street you are crossing (the audio quality though…)
- Has a locator tone to help find it
- Points in the direction of the crossing
- Often broken…and even worse it's not always obvious when they are broken.

Vision Challenges: Audio Wayfinding



Other common audio wayfindings

- Chirps and beeps at crossings
- Transit line and stop announcements
- Bus stop information

Vision Challenges: Tactile Wayfinding

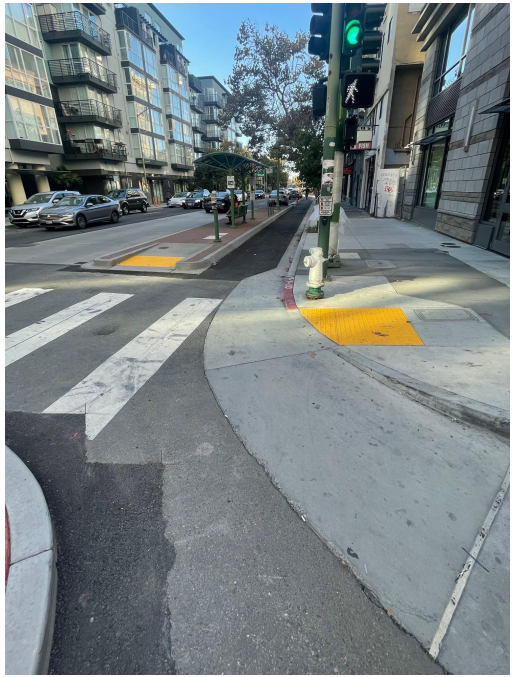


Truncated Domes/Tactile Pavers/Detectable Warning Surfaces

- All different names for the same thing
- *Usually* communicate a border between a safe pedestrian right of way and something that isn't that (a street, a massive pit where the train is gonna come, maybe a bike lane)
- Often the main way to give direction….
- Don't have quite enough standardization and/or widespread understanding as to what they mean

Vision Challenges: Tactile Wayfinding

A lot of practitioners don’t really know what they are doing…







Vision Challenges: Crosswalk alignment

A lot of practitioners don’t really know what they are doing…







Vision Challenges: Orientation



- Street signs are usually unreadable to blind folks.
  - Around here they appear to be hard to find and read for sighted people
- Low vision and blind folks have a hard time placing themselves in their wider environment
- Considerations like sidewalk stamping, done in SF allow many more people to know better where they are!
  - Wasn’t actually implemented as accessibility treatment, but as an emergency preparedness measure for earthquakes in 1905

Vision Challenges: Orientation



What is wrong with this sign?

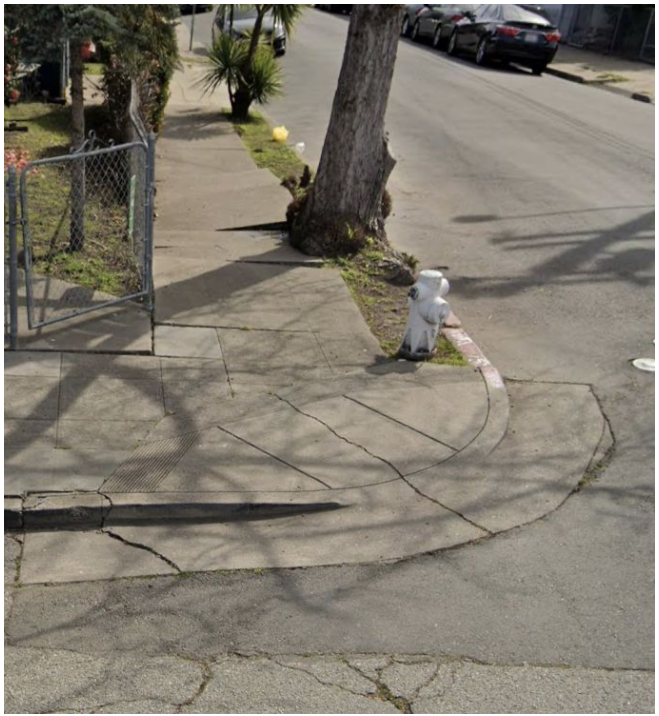
Vision Challenges: Orientation



What is wrong with this sign?

- Only 10% of blind people read braille
- The folks who do maintenance on these signs generally do not read braille
- We should be including raised text, so that all blind people, and lots of low vision and sighted people, can benefit.
- Hard to see, but this braille is actually **very** incorrect……

Vision challenges: Sidewalks

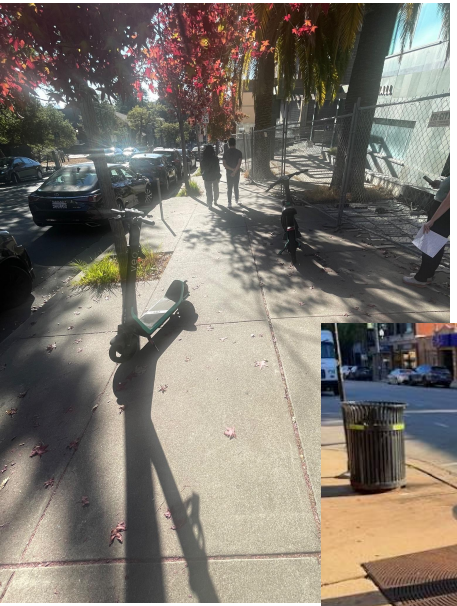


Uprooted, cracked, and otherwise damaged sidewalks are often a dead end for wheelchair users, **not to mention major trip hazards for everyone else**

Obstructions to the sidewalk can be structural damage, but often they are also trash or **cars parked on the sidewalk (!!)**

Blind folks are also affected by even planned and intentional infrastructure that clutter our sidewalks

Vision challenges: Obstructions





Scooters left in the middle of the sidewalk are a hazard for folks who aren't expecting them to be there, or folks who don't have space to move forward while they are there.

Up til now, we've mostly dealt with stationary obstructions…

Vision Challenges: Construction





Vision challenges: Open Spaces



Difficulty going straight and keeping directions can make places like parks and plazas very hard to navigate for blind folk.

Having intentional and detectable pathing for blind people can make these spaces less daunting, and more navigable.

Vision challenges: Open Spaces





Difficulty going straight and keeping directions can make places like parks and plazas very hard to navigate for blind folk.

Having intentional and detectable pathing for blind people can make these spaces less daunting, and more navigable.

How do mistakes happen?

Many of the shown errors are **very** apparent to people with lived experiences, properly consulting people with disabilities is crucial to good design

Well Intentioned, Poorly Informed: Bike Infrastructure



Paratransit and vehicles that serve wheelchair users need curb access, which is blocked here

Bike racks serve as additional obstacles for blind folks navigating a sidewalk

Well Intentioned, Poorly Informed: Sidewalk-level bike lanes



They **can** solve the problem of losing curb access for wheelchair users, but when they have greenspace between the bike lane and the car lane you lose that.

The separation for blind folks just disappears for long stretches

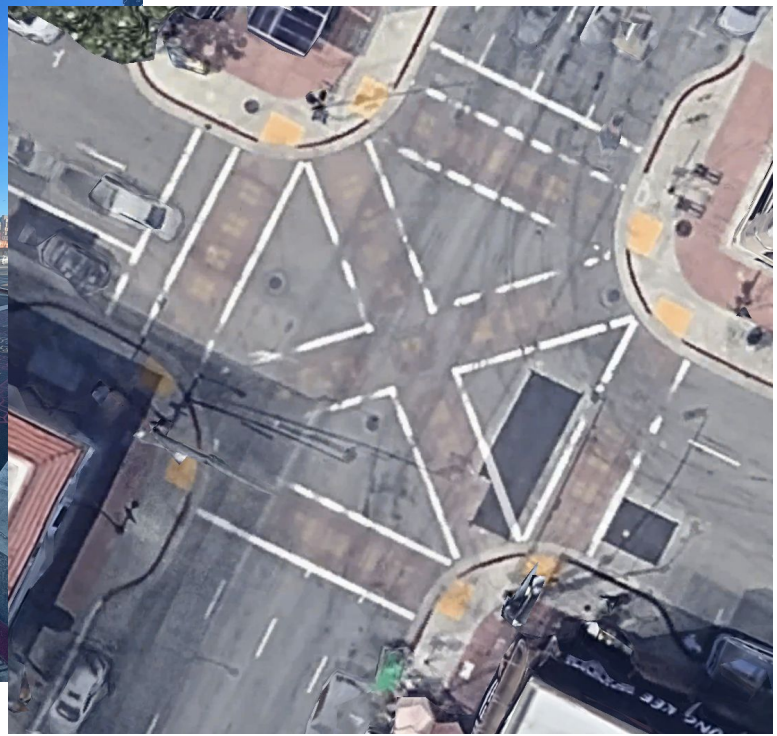
Well Intentioned, Poorly Informed: Sidewalk-level bike lanes



Even when the separation for blind folks exists, people misunderstand how people use it

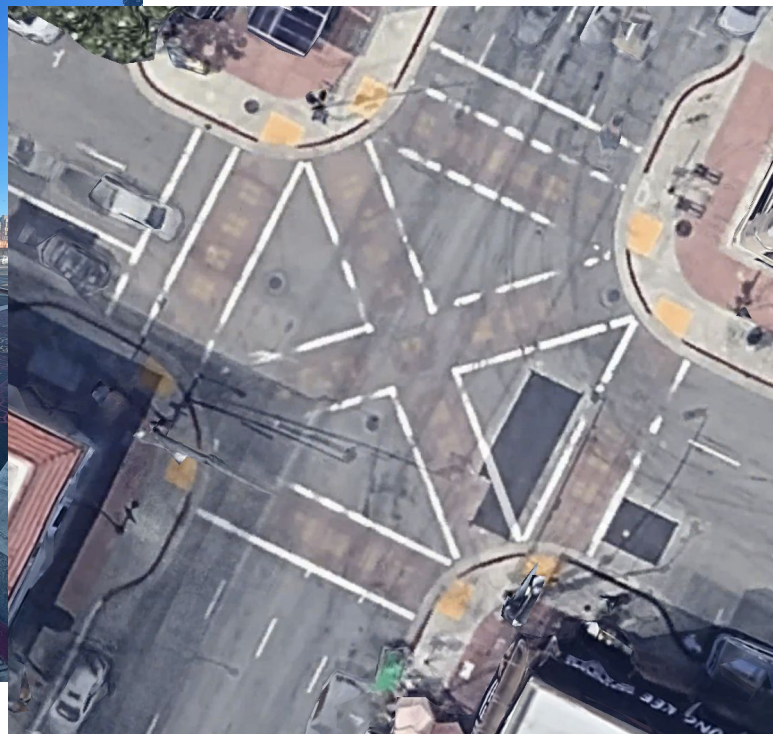
Well Intentioned, Poorly Informed: Pedestrian Scrambles





Well Intentioned, Poorly Informed: Pedestrian Scrambles





Much safer for pedestrians to cross, but no clear indication for blind people

The cues blind people often use, the sound and direction of cars for example, are absent here

Next Step, Common themes

With lots of qualitative observations made, we can start to think about what kinds of problems we want to work on, and only then can we start to think about quantitative methods and data

Next Step, Common themes

What stand out as important recurring issues across both mobility and vision?

Next Step, Common themes

What stand out as important recurring issues across both mobility and vision?

For me,

- Sidewalks, existence, state, and obstruction status
- Curb Ramps, type, directionality
- Construction locations and obstruction status

Before we can talk about what to do with these observations…

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What Data would we like? What do we have?

It would be great to have a detailed map of all the sidewalks, ramps, and construction sites, and their conditions

With that information, if we want to fix them, then knowing how people, pedestrians specifically, travel–that is where and how much–might in combination with equity factors help us prioritize what locations get earlier treatments.

What Data would we like? **What do we have**?

Sidewalk studies are very **slow and expensive**

Tools to scalably count people at specific locations–**especially without major privacy violations**–don't really exist or aren't used

Before we continue, a detour to help address the quality of data we want…

Notable risk of data: Aggregation

What is Aggregation? Combining data for various reasons such as general accuracy or generalizability.

E.g. Census tract-level data have high margins of error, so they are often combined for a better sample size,

e.g. Walk Scores or access ratios summarize complex data to give a simple score

Leads to the erasure of important details, including information about people with disabilities

We don’t just want # of ramps per neighborhood, we want to know where they are

We don’t just want a walk score, we need to know if the amenities counted make up the full set of services a person needs

Notable risk of data: Aggregation

High Injury Network in Chinatown



Top 3 intersections for injury collisions (2013-2023):

Harrison St & 12th St: 51 crashes

Harrison St & 14th St: 41 crashes

Webster St & 7th St: 34 crashes

Notable risk of data: Privacy and Security

Location or count data, such as that which an entity like Google or cellphone data might supply, are often so detailed as to easily deanonymize subjects, and at high risk of targeting by data breaches or other dangerous behaviour.

People with disabilities especially are vulnerable targets whose data needs to be treated with the utmost of care and competence.

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What Data would we like? **What do we have**?

As it happens, there are new tools being developed that may help with questions like these, and by none other than our own professor here!

Using computer vision to detect and map sidewalks and do geographically focused and anonymous traffic counts.

There are limits to what Computer vision can detect in the realm of accessibility even as we've discussed today

Methodology

Construct network graphs: Navigate the transition from GIS project to Pythonic data structures and graphs

Weight Edges: Write a weighting scheme using facility type, distance, and safety conditions

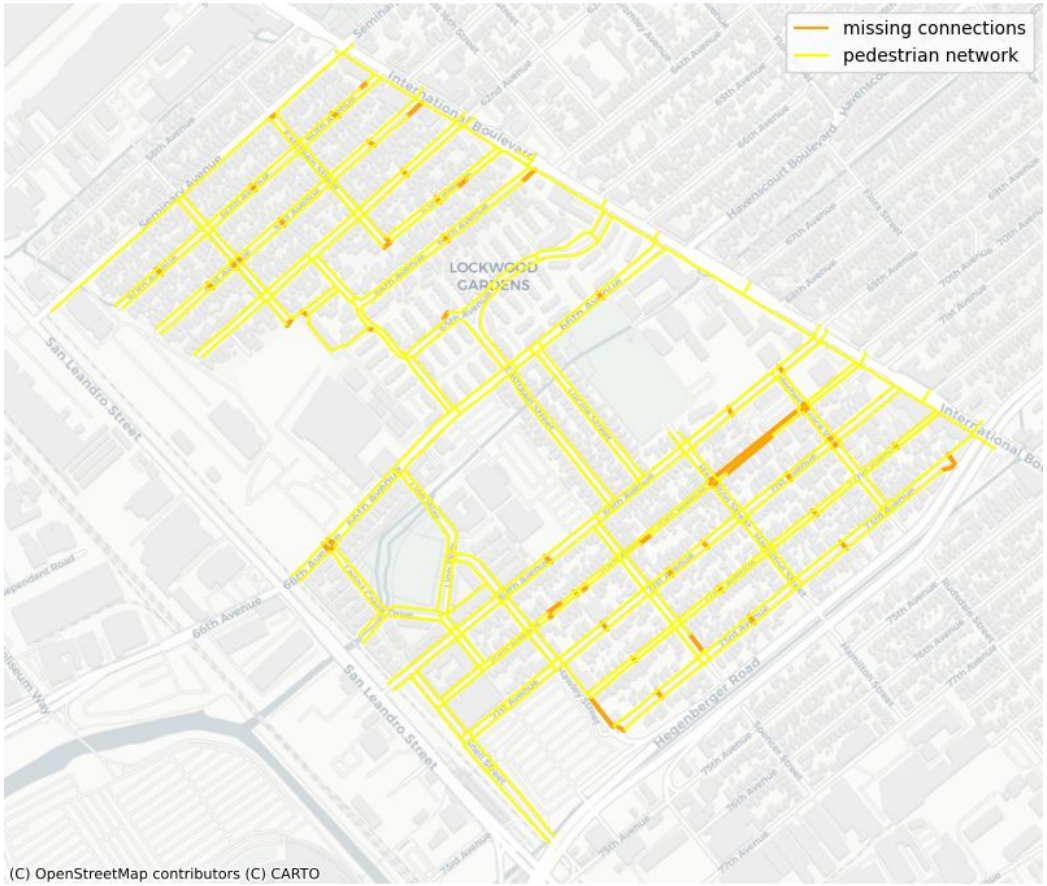
Run Networkx Shortest Path: Running Dijkstra's shortest path algorithm to map routes before and after gap closures

Prioritize Gaps: Extract the gaps most important to new routes to prioritize them



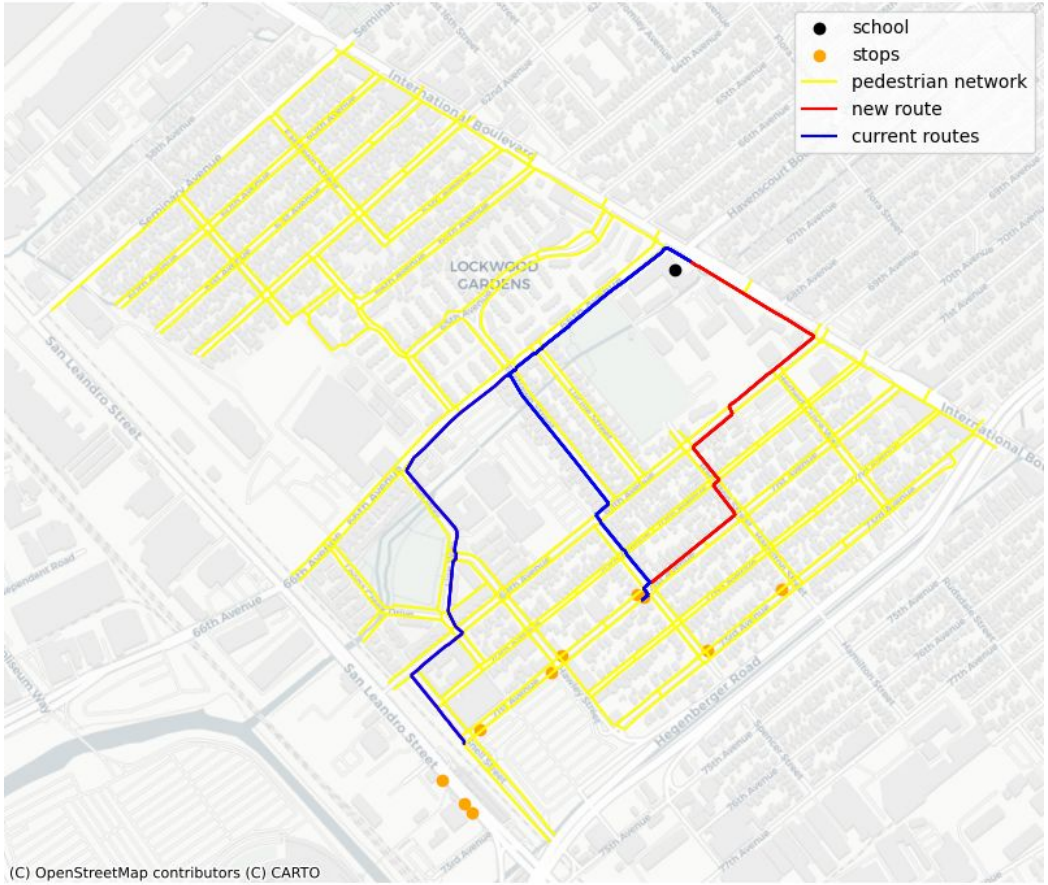
EPC Shortest-Path Analysis and Maps

Oakland Airport



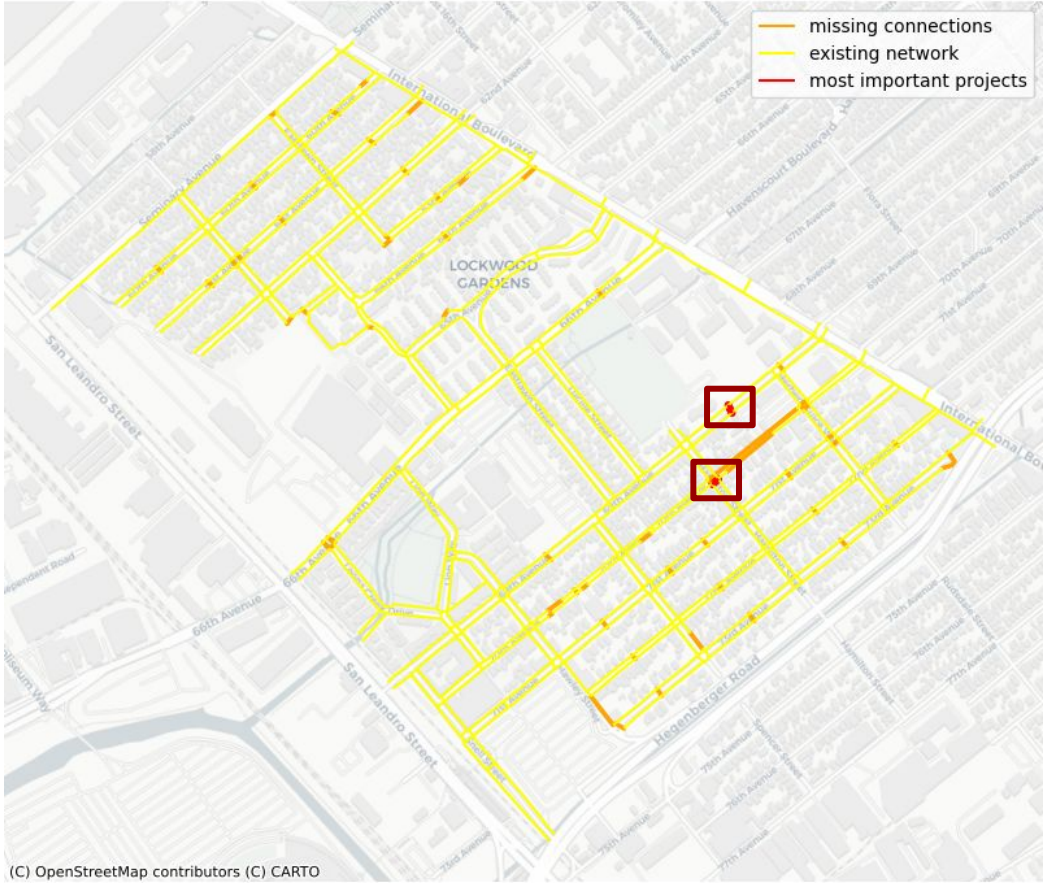
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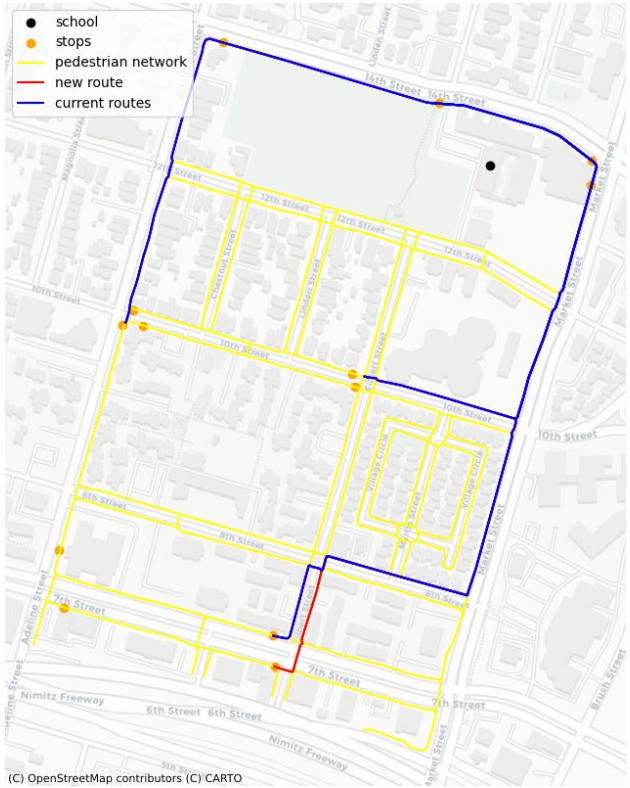
EPC Shortest-Path Analysis and Maps

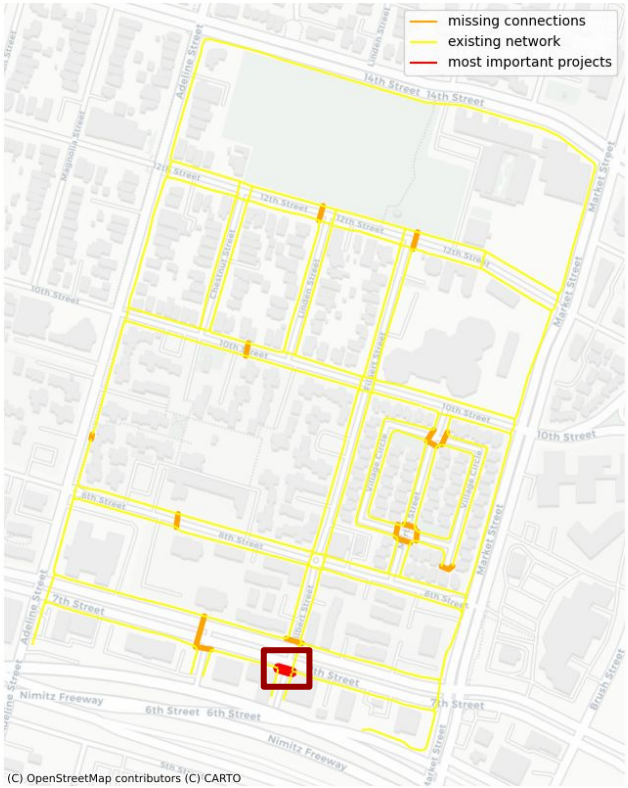
Oakland Airport



EPC Shortest-Path Analysis and Maps

West Oakland





Center and Shattuck:



[google earth link]

We can see:

We can’t see:

Center and Shattuck:



[google earth link]

We can see where curb ramps and paths of travel are, we can see sidewalk widths, and even some crosswalk misalignment.

We can’t see where push buttons are or if they are working, and may not be able to detect subtle obstructions

Exercise

Y’all compiled a list of places earlier this semester, try to use a map to chart **accessible** routes from campus, or from transit, to those places

Afterwards, go there in person. Are there things on the ground that your electronic observations couldn’t catch?

Reporting on both of those will help us understand the capabilities and limits of computer vision on tasks like these

The ability to, using satellite imagery, map accessible routes, and denote inaccessible ones for treatment, at a very low cost, could be exciting!

Final Key Takeaways

You have been imparted with new vision; you will start to see wheelchair infrastructure and crosswalk misalignments, feel sidewalk obstructions, hear locator tones and crossing beeps/chirps, and start to recognize massive and detrimental gaps and flaws in the accessibility of our infrastructure. Channel any new rage responsibly, and spread that vision to others

Quantitative data/methods has its limits, there is so much essential information captured by qualitative methods; don’t underestimate them

Designing for people with disabilities is going to impact way more people than you would think

Consulting with people with disabilities is **essential**

Thanks!

I’m happy to take questions for as much time as we have left

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